

DELIVERABLE # 1

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CENG-322-OCC, Software Project/Hardware Production

Deliverable 1

FINAL DOCUMENT LAYOUT

Students who are not part of a team, will get 0.

Please submit a single PDF document per group. This means that the document should be professionally organized and have a uniform style throughout. It should look as though it came from one team, not 4 separate students. Please note that instructors may choose to run your submission through TurnItIn or compare the submission with other students from other sections for academic integrity purposes.

A single PDF document with the following sections denoted using page numbers, headers/footers and a table of contents:

Must follow the sequence below:

Use proper formatting

1. Document Title: **Deliverable 1**

2. Team Name **LIFE (*Light, Indoor air, Freshness, and Energy*)**

3. Project Name: **LIFE Environmental Solutions**

4. Team Number:**11**

5. Student's names and IDs: Mohamed Ali **Student ID: N01440760**, Jonathan Akabuike **Student ID: N01510573**, Kieran Sharma **Student ID: N01548225**, Farhan Habibzai **Student ID: N01610299**

6. Table of Contents

Title: LIFE Environmental Solutions

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1. Project Background and Description

Project Goals and Final Vision

- **Software and Hardware Description**
- **Screen Flows**
- **Incorporated Feedback**
- **Database Read/Write Demonstration**

2. Project Scope

- **Project Plan and Completion Criteria**

3. Software-Hardware Integration

- **Integration of Hardware Components**

4. Project Layout

- **Navigation Drawer Implementation**
- **Bottom Navigation Implementation**

5. Project Theme: LIFE Environmental Solutions

- **Epic 1: Reliable Environmental and Human Presence Data Capture**
 - **Story 1: Human Presence Sensor Integration**
 - **Task: Connect sensor to microcontroller**
 - **Task: Write code to detect and count humans**
 - **Task: Send presence data to cloud database**
 - **Story 2: CO₂ and Humidity Sensor Integration**
 - **Task: Wire SGP30 and humidity sensor**
 - **Task: Write firmware to collect readings**
 - **Task: Push sensor data to database**
 - **Story 3: INA219 Current/Power Sensor Integration**
 - **Task: Connect INA219 module via I2C**
 - **Task: Write firmware to calculate power consumption**
 - **Task: Push power usage data to a cloud database**
- **Epic 2: Software and Cloud Integration**
 - **Story 1: Android App Development for Real-Time Monitoring**
 - **Task: Create app layout in Android Studio**
 - **Task: Implement API calls to fetch data**
 - **Task: Display graphs and indicators**
 - **Story 2: Cloud Database and API Setup**
 - **Task: Deploy cloud database (Firebase)**
 - **Task: Build API for read/write operations**
 - **Task: Test database read/write functionality**
 - **Story 3: User Authentication and Notifications**
 - **Task: Implement Google login integration**
 - **Task: Set up push notifications**
 - **Task: Test notification and authentication workflows**

7. Team Contract (from CENG-322_TEAM_contract_v2.0) **(DONE)**

8. Signature of each student who participated in the work (team

members who did not participate and where their signature is missing will get 0.)

9. In the pdf, write the requirement and comment on it for each item. Include the question and the answer.

10. Create Google Play Developer Account, take a screenshot showing account has been approved and created (one or more members).

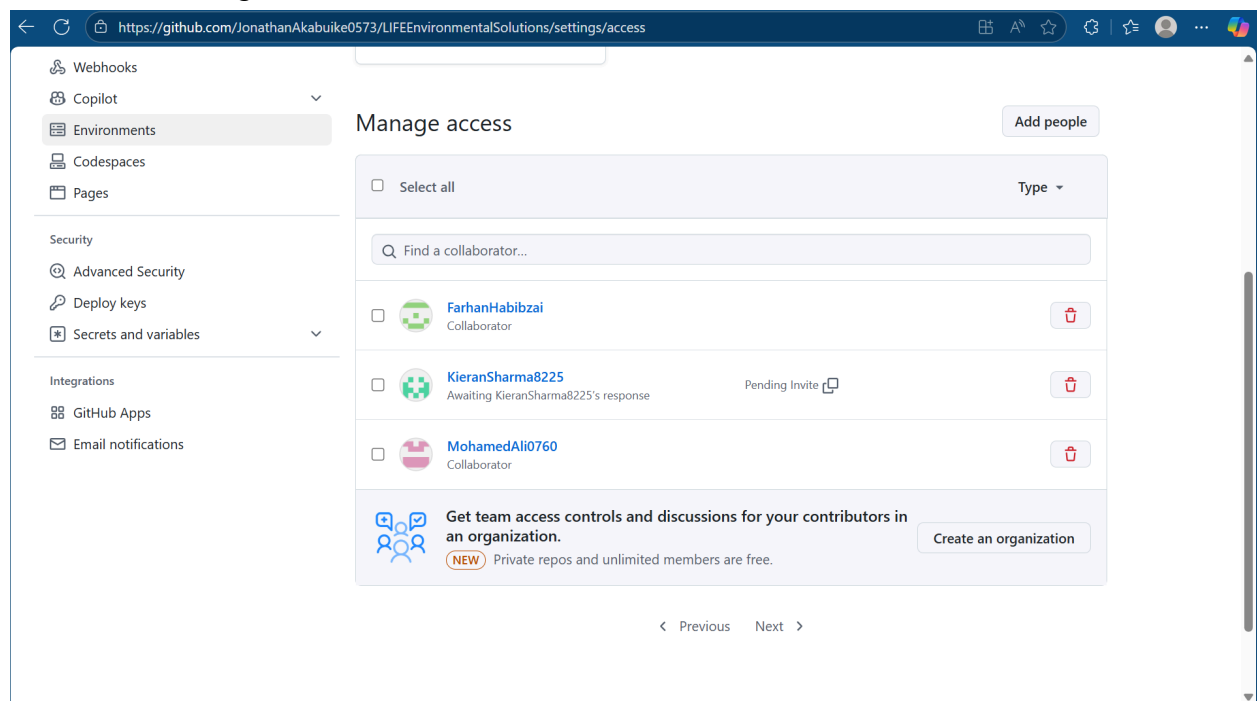
<https://support.google.com/googleplay/android-developer/answer/6112435?hl=en#zippy=>

11. Link GitHub Repo. All members must contribute to the repo. **(CURRENTLY WORKING ON IT)**

12. Each member must have a minimum of three commits.

Missing commits will get 0. Commits must be done from Android Studio.

13. Screenshot of github invitation.



14. Project Background and Description: **Our project focuses on creating a smart system that monitors and improves indoor environmental conditions for comfort, safety, and energy in spaces (e.g classrooms, offices, and homes.) C02, humidity, saving energy and checking the number of people in a space using a Human Presence sensor that uses Infrared.**

1. Describe the project goals and final vision.**We believe our project addresses a real and relevant problem because poor indoor environmental conditions can directly affect comfort, health, and energy efficiency. Many people spend most of their time indoors, whether in warehouses, classrooms, offices, or homes, and issues such as stale air, insufficient lighting, or wasted energy often go unnoticed.**

2. Describe the software aspect and hardware.: **The hardware aspect will have the sensors working to sense human presence, air quality and can potentially have a way of saving energy. The Software will have java code integrating it from android studio will a google play account part of the project later on.**
3. Describe the screen flows.: **The screen flows will provide the setup/layout of our project and will show what is to be done and the intention is for it to be what we want our hardware and software to be once the project is completed.**
4. How you incorporated the feedback provided through the interview.: **We incorporated the feedback from the interview by now adding onto our main goal a humidity and CO2 detection. We also will now have the Human Presence Sensor now be able to sense multiple humans using an infrared system.**
- 14.5. Demonstrate how you are planning to satisfy to read / write from the DB which is hosted on the cloud.: **We will be satisfying the read/write from the DB by having the database connected library on the connections, having database operations done, implementing a database handler and having a connection library.**
15. Project Scope: Describe the technical scope of the project by talking about the project plan, and how you will know when the project is complete.: **We will know when the project will be completed when we have accomplished our goals in getting the hardware and software to link up, to show the readings that include the carbon dioxide and humidity reads, as well as the human presence.**
16. Describe how to integrate the software with the hardware components.**The hardware will integrate multiple hardware components and it will demonstrate how they interact to solve real-world problems. We will be using: Human Presence Sensor, INA219 I2C Bi-Directional DC Current Power Supply Sensor Breakout Module 2PCS To have a smart system that monitors and improves indoor environmental conditions that includes our comfort, safety, and energy. It helps in the saving energy dept. and SGP30 Air Quality Sensor to collect physical data from the environment. These sensors will be key in having the physical part of our project. The software will allow us to integrate the hardware by coding from Android Studio, and having applications from google play that will help us in monitoring the system.**
17. Project layout: write how you plan to implement Navigation Drawer, and combination with another layout. **The Navigation Drawer will act as the primary navigation hub of our project. It will provide quick access to all major sections and will have a clear and consistent user experience regardless of the user's current location within the app.**

The drawer will be accessible from the top-left corner of the screen via a hamburger menu icon. Contains the following menu items:

Dashboard: This screen is our home screen, a summarization of all monitored spaces.

My Spaces: A list containing all rooms or locations, allowing the user to select one to view the data.

Analytics: A section where viewing historical data and trends through charts and graphs.

Energy Tips: A section where the user will find tips on saving energy

Settings: The configuration screen for the app.

Our next layout will be the bottom navigation. This navigation will allow us to have the ability to have a presence and a highly visible control panel. Unlike the navigation drawer which is hidden, the bottom navigation is there for us to be easily accessed and there is a quick switch for the user if the user wants to change from one screen to another screen.

The combination of both a Navigation Drawer and a bottom navigation bar creates a robust and intuitive navigation system. Here is the proposed strategy for their coexistence:

Bottom Navigation Bar (Primary Navigation): This will house the four or five most critical destinations of our app. For our project, this would likely include:

Dashboard: The main screen for the real-time data

My Spaces: A list of all monitored rooms.

Analytics: Detailed charts and historical data.

Settings/Profile: A quick way to access user preferences.

<https://m2.material.io/components/navigation-drawer/>

<https://m2.material.io/components/bottom-navigation>

<https://m2.material.io/components/tabs/android>

How to use them, why to use, use cases of existing applications that utilize them. one vs the others, and use cases of existing apps that use the above.

18. Write Minimum of 1 theme: for the Theme write: 1. Two epics, 3 stories for each epic, 3 tasks for each story. Theme: The theme is LIFE Environmental Solutions, A system that enhances comfort, safety, and energy efficiency by monitoring CO₂, humidity, light, and human presence in indoor spaces. Epic 1: The first epic is to be able to capture and have reliable environmental and human presence data by using multiple sensors we have chosen and to use them in our project. The first story will be how we integrate our human

presence sensor. The tasks are: Connect Human Presence sensor to microcontroller, Write down java code to detect and count out multiple humans in a room and finally, Send presence data to cloud database. The second story of epic 1 is to Integrate the CO₂ and Humidity Sensor into the project. The tasks for these are: to have the Wiring of the SGP30 and humidity sensor to the hardware platform. To Write firmware to collect CO₂ and humidity readings. Finally, to Format and push sensor data to the database in real-time. The last story in epic 1 is to Integrate the INA219 Current/Power Sensor. The tasks are: to connect the INA219 module via I2C to measure current and voltage from the power supply. To Write firmware to calculate power consumption ($P = V \times I$) from THE INA219 readings. Finally, to push power usage data to a cloud database for energy efficiency tracking and energy saving. Epic 2: The second epic is to have software and cloud integration into our project. The first story of epic 2 is to Develop an Android App for a Real-Time Monitoring system. The tasks are: to create an app layout in Android Studio (Java is the language we use). We will implement API calls to fetch data from cloud DB. and finally, to have a chance to display graphs/indicators for CO₂, humidity, light, and occupancy. The second story of epic 2 is to Set Up Cloud Database & API. The Tasks are: to Deploy cloud DB (Firebase is the DB we use). To Build | API for read/write operations. And finally, to test the DB so that it can read/write functionality with sample sensor data. The last story of epic 2 is to have User Authentication & Notifications. The tasks are: to Implement Google login integration (Google Play services). Set up push notifications (e.g., air quality exceeds threshold). And finally, test the notification and authentication workflows, to see if it works.

19. If document not in pdf, or not the proper name (marks will be deducted!)

20. Please save the PDF document, i.e.

TeamName_ProjectName_GroupNumber_Deliverable_1.pdf, and submit on the blackboard.

21. The last submitted pdf file will be marked.

Android Studio Project pushed into in github:

22. Please Update your github Id, to be YourFullName4Digits, i.e. JohnSmith1111.

23. Replace John with your first name, Smith with your last name, and 11111 with your studentId.

24. Use Navigation Drawer with the combination with another layout.

25. Minimum API 30.

26. Target API 36/37

27. Create a new project, File -> New -> New Project.

28. Project name must be YourProjectName

29. Module name must be YourModuleName.

30. Package name must be ca.teamname.it.projectname (i.e. ca.security.soluHons.it.home.security)
 31. Java classes must start with upper case.
 32. Use a proper name for your Java classes that reflects the functionality.
 33. All resources in strings.xml, no hardcoding of text.
 34. Must Support English and French.
 35. Full Name, student ID, and secHon number, must be on ALL JAVA classes as a comment.
Name
and student id of all participants.
 36. Full Name, student ID, and secHon number, must be on both strings.xml as a comment.
Name
and student id of all participants.
 37. In strings.xml, app_name must be your project name.
 38. MUST use Fragments for the individual screens.
 39. Demonstrate the use of styles in objects.
 40. Use a homescreen icon different from the default Android icon. Use an icon that reflects the app
behaviour.
 41. Intercept user press on back key and display Alert Dialog whether user wants to stay on app
or
exit. Use a proper icon for the dialog.
 42. Every Home you want to commit, select root directory, Commit using Git -> Commit
Directory.
 43. ApplicaHon must not have compilaHon errors.
 44. Application must not crash.
 45. Implement Splash Screen when the user clicks on the homescreen icon, and then land the
user on the Main Screen.
 1. Use implementation libs.core.splashscreen as demonstrated in the class
 2. Splash screen appears for 3 seconds, then the user lands on the main screen.
 3. Design of two splash screens, one for default theme and one for night theme.
 Implement some type of animation on the splash screen.
- [hCps://www.youtube.com/watch?v=eFZmMSm1G1c&ab_channel=PhilippLackner](https://www.youtube.com/watch?v=eFZmMSm1G1c&ab_channel=PhilippLackner)
- Github Requirements: Follow instructions below**
- Repository name must be YourProjectName, no spaces.
46. Make your repository private (not public).
 47. One of the team members to invite the professor as a collaborator (invite haki11). Take a screenshot of the invita3on. Screenshot must show the other team members and the professor.
 48. Must Commit on your machine and push into github.
 49. ARer each commit, push.

50. Each member must have a minimum of 3 commit/pushes into master branch.
51. Every time you want to commit, select root directory, Commit using Git -> Commit Directory. The entire project must be commi[ed and not just the module.
52. Add .gitignore file with the appropriate content into the project folder. Add team name as a comment into the first line of .gitignore (Then add all members names and students ids).
[h\[ps://www.toptal.com/developers/gitignore](https://www.toptal.com/developers/gitignore)
Please note the file .gitignore must be at project level and not module level (Delete the one in your module).
53. Add README.md file into the project folder, with the appropriate content explaining about your project. Use the appropriate tagging format. Add Project name as first line in the file README.md
54. Add members names and students ids into the file README.
<https://www.makeareadme.com/>
55. Include project image in your README as in the example:
[hgps://github.com/johncodeos-blog/CheckInternetAndroidExample](https://github.com/johncodeos-blog/CheckInternetAndroidExample)
56. All members names and student's ids as second line.
[h\[ps://www.makeareadme.com/](https://www.makeareadme.com/)
57. Add project link into the README file.
Please note the file must be at project level and not module level.
- 58. At the project level, create a directory called docs, under docs create deliverable sub-folder. Copy the pdf into the sub-folder. Commit and push.**
59. The entire project should be committed and pushed where it can be cloned.
60. Your submission must be a single pdf file as described in step 20.